

redesigning. As a result, the development process tends to be more linear and predictable and, consequently, the time to market is reduced.

In short, the advantages can be summarised as: (a) an ability to test leading-edge functionality where there is no alternative; (b) enhanced ability to optimise the design for superior performance; (c) a lower risk of design problems being discovered late, causing costly rework or delays; and (d) reduced risk of financial losses or reputational damage.

Challenges

Most leading consumer and industrial electronics manufacturers already use this methodology, because they understand that the costs and risks of not doing so are too high. Also, if they want to be ahead of the competition, they cannot wait for technology to mature—so they need to test during design. However, for companies working on mature technologies, some regard

the costs of investing in additional test equipment early in the design stage as too high. Stephen Hire agrees, "It is easy to quantify the costs of extra test equipment but many engineers forget to factor in the costs of product launch delays, excessive customer returns or damage to a company's reputation from inadequately tested products. Let's take an example of a product with a selling price of ₹ 5000 and an expected monthly sales volume of 10,000 units. Every one week of delay for the product launch will cost the company approximately ₹ 12,500,000 in lost sales revenue; and that doesn't even account for the cost of faulty products returned or the damage to reputation. A short-cut in the design stage can quickly add up to a huge financial damage."

A typical challenge of testing during design is of simulating the conditions of the overall system during lab conditions. "In order to simulate the right conditions, it typically calls for extra effort in writing test cases and test programs,

which some people might consider as an overhead during design. And because of this, some engineers might refrain from doing so," adds Amit Sinha.

In general, the time-to-market pressure is the biggest factor. There is also a lot of pressure to see the entire system working. Unenlightened management might see an involved process of thoroughly testing subsystems as a waste of time. According to Teledyne Lecroy spokesman, "There is often a conflict between testing at the system and subsystem levels, and achieving a balance between them takes much thought and planning. Designers who are trained primarily in design alone must think about the testing as much as the design. More experienced engineers do a better job of factoring in testing, because they have suffered the consequences of ignoring it. Engineering schools do not teach their students to approach design with testing in mind. Rather, it is learnt with time and experience." ●

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